

Question				Marking details	Marks available						
					AO1	AO2	AO3	Total	Maths	Prac	
9.	(a)			Indicative content - percentage of chlorine in this compound is 32.1% which is consistent with that of 2,4-D - colourless gas with NaHCO_3 indicates compound is a carboxylic acid $n(\text{NaOH})$ is $1.85 \times 10^{-3} \text{ mol}$ - mole ratio is 1:1 therefore only one COOH group - number of moles of compound in 25.0 cm^3 is also $1.85 \times 10^{-3} \text{ mol}$ therefore 1.85×10^{-2} in 250 cm^3 M_r of compound is 221 which is consistent with that of 2,4-D test results consistent with structure of 2,4-D - no chloride ions on reflux with NaOH therefore chlorine atoms are bonded directly to the ring, not in alkyl side-chains - no white precipitate with aqueous bromine therefore not a phenol - aqueous bromine not decolourised therefore no $\text{C}=\text{C}$ double bonds spectral data consistent with that of 2,4-D ^1H NMR spectrum of this compound would show three peaks; aromatic (area of 3), $-\text{CH}_2-$ (area 2), $\text{O}-\text{H}$ (area 1) ^{13}C NMR of this compound would show eight peaks as it has 8 different carbon environments - take its melting temperature and compare to a book value - mix a sample with some actual 2,4-D, and take its melting temperature – see if the value is unchanged							
					2	2	2	6	2	4	